

JUHANA SILJANDER

www.juhanasiljander.com

j.siljander@imperial.ac.uk

Office Contact Information

Department of Finance

Imperial College London Business School

53 Prince's Gate, SW7 2AZ, London, UK

Office and/or cell phone number: +44 79 089 61867

Education

Ph.D. in Economics, University of Chicago

D.Sc. in Technology (mathematics), Aalto University

M.S. in Mathematics, New York University

M.Sc. in Mathematics, University of Oulu

References:

Professor Tarun Ramadorai

Imperial College London Business School

jmletters@imperial.ac.uk

Professor Joseph Vavra

Univ. of Chicago Booth School of Business

joseph.vavra@chicagobooth.edu

Professor Veronica Guerrieri

Univ. of Chicago Booth School of Business

erzsebet.vincent@chicagobooth.edu

Research Fields

Primary fields: Real Estate Finance, Household Finance

Secondary fields: Macro Finance, Macroeconomics

Teaching Experience

Instructor / Module leader

Autumn, 2022 Investments and Portfolio Management, Imperial College London

Spring, 2021 Economic Policy Analysis (advanced undergraduate macro), U of Chicago

Autumn, 2019 Economic Policy Analysis, University of Chicago

Summer, 2016 Differential Equations, University of Jyväskylä

Teaching Assistant

Autumn, 2023 Principles of Finance, Imperial College London

Autumn, 2019 Macroeconomics (MBA), Chicago Booth

Summer, 2019 Macroeconomics (MBA), Chicago Booth

Spring, 2019 Economic Policy Analysis, University of Chicago

Winter, 2019 Investments (MBA), Chicago Booth

Autumn, 2018 Macroeconomics (MBA), Chicago Booth

Spring, 2010 Basic Course in Mathematics L4 (Honors PDEs), Aalto University

Spring, 2009 Basic Course in Mathematics L4, Helsinki University of Technology

Autumn, 2008 Calculus I, New York University

Employment

2022 – 2025	Imperial College Business School, postdoctoral research associate (finance)
2015 – 2016	University of Jyväskylä, postdoctoral researcher (mathematics)
2012 – 2015	University of Helsinki, Academy of Finland postdoctoral researcher (mathematics)
2012	University of Oulu, postdoctoral researcher (mathematics)
2010 – 2012	The Boston Consulting Group, Associate

Honors, Scholarships, and Fellowships

2021	Yrjö Jahnsson Foundation Graduate Grant
2020	Data Acquisition Grant, University of Chicago Dept. of Economics
2017	CRSP Summer Research Grant, Booth School of Business
2016 – 2022	Social Science Graduate Fellowship, University of Chicago
2014 – 2016	Väisälä Foundation Travel Grants
2012 – 2015	Academy of Finland postdoctoral research fellowship in mathematics
2007 – 2009	Fulbright Foundation Scholarship
2007	Alfred Kordelin foundation Graduate Scholarship
2006	Ernst Lindelöf prize (master's thesis award), The Finnish Mathematical Society

Professional Activities

Conference and Seminar Presentations (economics and finance):

2025	AREUEA-ASSA (scheduled)
2024	Urban Economics Association European Meeting, Normac Junior Macro Conference, Helsinki Finance Summit, Bank of Finland, Swansea, LSE Hellenic Observatory (scheduled)
2021	Young Economist Symposium (Princeton), 16 th Annual Economics Graduate Student Conference (WUSTL), UChicago Joint Program and Friends Conference

Referee (economics and finance):

Review of Financial Studies

Referee (mathematics):

Journal de Mathématiques Pures et Appliquées, Journal of Differential Equations, Nonlinear Analysis: Theory, Methods & Applications, Manuscripta mathematica, Annales Academiæ Scientiarum Fennicæ Mathematica, Journal of Mathematical Analysis and Applications, Applied Mathematics Letters

Language and Computer Skills

Computer Skills: R, Matlab, Dynare, Mathematica

Languages: English (fluent), Finnish (native)

Working Papers:

“Behavioral Lock-in: Aggregate Implications of Reference Dependence in the Housing Market” with Cristian Badarinza, Tarun Ramadorai, and Jagdish Tripathy (First Job Market Paper)

Abstract: Households' attachment to nominal anchors in the housing market creates aggregate nominal rigidities, with material economic consequences. A new statistic, the prevalence of "paper losses" in the stock of residential properties, captures cross-regional variation in important housing market quantities;

the share of properties facing paper losses explains housing transactions volumes more effectively than price growth rates. To rationalize these patterns and study housing-market fiscal policy, we develop and structurally estimate a dynamic search and matching model with reference-dependent agents, rich heterogeneity, and realistic financial constraints. Behavioral frictions dampen the tax elasticity of property prices, and increase the revenue-maximizing level of property taxes.

“Entrepreneurs and Housing Boom-Bust Dynamics” (Second Job Market Paper)

Abstract: This paper builds a dynamic general equilibrium housing model to propose a new mechanism for explaining how credit supply shocks can generate housing boom-bust episodes. The model highlights how investors' capital allocation links to real estate dynamics, focusing on credit-constrained firms rather than credit-driven household demand. The model embeds a credit cycle, which generates an endogenous bust without an explicit shock reversal. Additionally, the model explains several other typical housing macro dynamics, such as house price momentum, large consumption elasticities out of housing wealth, and decreasing interest rates during a sustained house price boom. Quantitatively, the model can match the observed U.S. house price increase during the early 2000s.

“Entrepreneurs, Collateral Channel, and the House Price Volatility Puzzle” (submitted)

Abstract: This paper proposes a new theoretical mechanism for explaining how a relaxation of entrepreneurial collateral constraints on business capital can increase house prices disproportionately. Such a credit relaxation enables productive entrepreneurs to purchase business capital, triggering capital reallocation and increasing aggregate productivity. Extending the Evans and Jovanovic (1989) model of entrepreneurial financing with housing markets, I demonstrate how this may produce a housing boom through general equilibrium effects. Empirical analysis leveraging Community Reinvestment Act (CRA) business loan data suggests that the mechanism can plausibly explain a notable portion of the pre-crisis increase in U.S. house prices during the early 2000s.

“Talent allocation, aggregate productivity, and income inequality: Evidence from Finland”

Abstract: This paper studies the impact of occupational talent allocation on aggregate productivity and income inequality. I use Finnish administrative microdata to estimate a Roy model of occupational choice with unobservable skill and preference heterogeneity. I show that workers' sorting behavior changes have not driven aggregate productivity growth in the recent past. Holding skill distribution fixed, potential future gains for aggregate productivity by improving sorting are also limited. However, different time trends on occupational sorting patterns explain up to 40% of relative wage growth in certain occupations during 1995 – 2005. In particular, differential occupational sorting between genders explains 3.5 percentage points, or 16%, of the gender earnings gap. When accompanied by changes in the skill distribution, workers' occupational sorting behavior also matters for aggregate output. Removing gender differences in skills, in particular, would lead to a 28% higher GDP effect than complete gender equalization with identical sorting patterns. I augment this analysis by leveraging the staggered implementation of the *Finnish Comprehensive School Reform* to discipline another counterfactual exercise. I use the model to decompose the reform effect into its skill and sorting components. I show that the reform's differential impact on women increased aggregate productivity by one percent, half of which was due to the sorting channel.

Selected publications (mathematics)

On the parabolic Harnack inequality for non-local diffusion equations (with D. Dier, J. Kemppainen, and R. Zacher). *Math Z.* 295 (2020), 1751–1769.

Boundary regularity for the porous medium equation (with A. Björn, J. Björn, and U. Gianazza). *Arch. Rational Mech. Anal.*, 230 (2018), no. 2, 493–538.

On the interior regularity of weak solutions to the 2-D incompressible Euler equations (with J.M. Urbano). *Calc. Var. Partial Differential Equations*, 56, 126 (2017).

Representation of solutions and large-time behavior for fully nonlocal diffusion equations (with J. Kemppainen and R. Zacher). *J. Differential Equations*, 263 (2017), no. 1, 149–201.

Everywhere differentiability of viscosity solutions to a class of Aronsson's equations (with C. Wang and Y. Zhou). *Ann. Inst. H. Poincaré Anal. Non Linéaire*, 34 (2017), no. 1, 119–138.

Decay estimates for time-fractional and other non-local in time subdiffusion equations in $\mathbb{R}^d$ (with J. Kemppainen, V. Vergara, and R. Zacher). *Math. Ann.*, 366 (2016), no. 3–4, 941–979.

Hölder continuity for Trudinger's equation in measure spaces (with T. Kuusi, R. Laleoglu, and J.M. Urbano). *Calc. Var. Partial Differential Equations*, 45 (2012), no. 1–2, 193–229.

Local Hölder continuity for doubly nonlinear parabolic equations (with T. Kuusi and J.M. Urbano). *Indiana Univ. Math J.*, 61 (2012), no. 1, 399–430.

Obstacle problem for nonlinear parabolic equations (with R. Korte and T. Kuusi). *J. Differential Equations*, 246 (2009), no. 9, 3668–3680.